The Category Zoo

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Contents

1 Linear Categories **2**
- 1.1 Semiadditive category 2
- 1.2 Additive category 2
- 1.3 Preadditive category 3
- 1.4 $\Bbbk$-linear category or $\Bbbk$-category 3
- 1.5 Additive envelope 3
- 1.6 Karoubi envelope 3
- 1.7 Cauchy completion 4
- 1.8 Simple objects (non-abelian version) 4
- 1.9 Semisimple category (non-abelian version) 4

2 Limits and Colimits **5**
- 2.1 Kernel (via equalizer/pullback) 5
- 2.2 Cokernel (via Coequalizer) 5
- 2.3 Left/Right Exact Functors 5
- 2.4 Pullbacks from Products and Equalizers 6
- 2.5 Pushouts from Coproducts and Coequalizers 6
- 2.6 Ends and coends 7

3 Monoidal Categories and Related Structures **9**
- 3.1 Monoidal category 9
- 3.2 Left rigid category 10
- 3.3 Right rigid category 10
- 3.4 Rigid or autonomous category 10
- 3.5 Pivotal or sovereign category 10
- 3.6 Spherical category (classical definition) 10
- 3.7 Braided category 11
- 3.8 Ribbon or tortile category (take 1) 11
- 3.9 Balanced category 11
- 3.10 Ribbon or tortile category (take 2) 11
- 3.11 Monoidal $\Bbbk$-linear category 11
- 3.12 Pre-fusion category (non-abelian version) 11
- 3.13 Fusion category (non-abelian version) 12
- 3.14 Modular category (non-abelian, fusion version) 12

3.15 Drinfeld center . . . 12
3.16 Müger center . . . 13
3.17 Equivalent Characterizations of Modular Categories . . . 13
4 Abelian Categories and Related Structures . . . 14
4.1 Kernels and cokernels . . . 14
4.2 Monos and epis . . . 14
4.3 Abelian category . . . 14
4.4 Linear + object semisimple imply abelian . . . 15
4.5 Monoidal abelian category . . . 16
4.6 Monoidal abelian $\Bbbk$-linear category . . . 16
4.7 Projective and injective objects . . . 17
4.8 Semisimple category . . . 17
4.9 Semisimplicity vs. Projectivity in Abelian Categories . . . 18
4.10 Locally finite abelian $\Bbbk$-linear category . . . 19
4.11 Finite abelian $\Bbbk$-linear category . . . 19
4.12 Left/right exactness, adjointness and (dual) representability . . . 19
4.13 Deligne product . . . 20
4.14 Kelly product . . . 20
4.15 (Multi)tensor, fusion and ring categories . . . 21
4.16 The distinguished invertible object (take 1) . . . 21
4.17 Unimodular finite tensor category . . . 21
4.18 Indecomposable projective objects in a finite tensor category . . . 22
5 Module categories . . . 23
5.1 Definition . . . 23
5.2 $\mathcal{C}$-module functor . . . 24
5.3 Module category over a multitensor category . . . 24

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1 Linear Categories

1.1 Semiadditive category

A category with a zero object (i.e., both initial and terminal object; and therefore zero morphisms) and with finite products and coproducts in which the canonical map

$$\coprod_i X_i \to \prod_i X_i$$

is an isomorphism—hence finite biproducts exist, and these are typically called direct sums $\oplus$.

1.2 Additive category

Any preadditive category is naturally enriched over commutative monoids: given maps $f, g : X \to Y$, define $f + g$ as the composite:

$$X \to X \times X \cong X \amalg X \to Y \times Y \cong Y \amalg Y \to Y$$

where the first map is the diagonal map $(\mathrm{id}, \mathrm{id})$, the second map is $(f, 0) \amalg (0, g)$, and the third one is $\mathrm{id} \amalg \mathrm{id}$.

If such an enrichment extends to abelian groups, that is, if every map f has an inverse $-f$ for the previously defined sum, then the category is said to be an additive category.

Alternatively, additive means Ab-enriched plus finite products and coproducts exist (and because of the enrichment they are automatically isomorphic). Typically, the biproduct is represented by the direct sum sign $\oplus$.

A functor between additive categories is additive if it preserves the finite biproducts, i.e., $F(0) = 0$ and $F(X \oplus Y) \cong F(X) \oplus F(Y)$.

1.3 Preadditive category

A category enriched over abelian groups.

1.4 $\Bbbk$-linear category or $\Bbbk$-category

A category enriched over $\Bbbk$-modules, where $\Bbbk$ is a commutative ring with unit. A linear category might *not* be additive (finite biproducts might not exist).

A functor between $\Bbbk$-linear categories is linear if $F_{X,Y}$ between the hom's is a linear map. Linear functors are automatically additive, i.e., they preserve the biproducts.

1.5 Additive envelope

As mentioned before, a linear category might not have all finite biproducts. But it always embeds into a category that has them. The additive envelope $\mathsf{Add}(\mathcal{C})$ of a linear category $\mathcal{C}$ has objects formal direct sums $\bigoplus_i X_i$ of objects of $\mathcal{C}$ and arrows given by matrices of arrows in $\mathcal{C}$, with composition modelled by matrix multiplication. There is a canonical functor $\mathcal{C} \hookrightarrow \mathsf{Add}(\mathcal{C})$, and it is determined by the universal property that if $\mathcal{D}$ has all finite biproducts, then there is a bijection

$$\mathrm{LinFun}(\mathcal{C}, \mathcal{D}) \cong \mathrm{LinFun}(\mathsf{Add}(\mathcal{C}), \mathcal{D})$$

1.6 Karoubi envelope

Remember that an idempotent in a category is an endomorphism $f : X \to X$ of a certain object such that $f \circ f = f$. A splitting for an idempotent f is a pair of arrows $r : Y \to X$ and $s : Y \to X$ such that $s \circ r = f$ and $r \circ s = \mathrm{id}_Y$. Splittings for idempotents are unique up to unique isomorphism.

A category $\mathcal{C}$ is idempotent complete or Karoubian if all idempotents split. A linear category might have idempotents that do not split. But it always embeds into a category where they do. The Karoubian envelope or idempotent completion $\mathsf{Kar}(\mathcal{C})$ of a linear category $\mathcal{C}$ has objects idempotents $f : X \to X$ (more precisely pairs (X, f)).

An arrow between two idempotents (X, f) and (Y, g) is an arrow $h : X \to Y$ such that $h = g \circ h \circ f$, with composition taken from $\mathcal{C}$. There is a canonical functor $\mathcal{C} \hookrightarrow \mathsf{Kar}(\mathcal{C})$, taking X to (X, id), and it is determined by the universal property that if $\mathcal{D}$ is idempotent complete, then there is a bijection

$$\mathrm{LinFun}(\mathcal{C}, \mathcal{D}) \cong \mathrm{LinFun}(\mathsf{Kar}(\mathcal{C}), \mathcal{D})$$

1.7 Cauchy completion

A linear category is Cauchy complete if it has all finite biproducts and it is idempotent complete. The Cauchy completion of $\mathcal{C}$ is $\mathsf{Cauchy}(\mathcal{C}) := \mathsf{Kar}(\mathsf{Add}(\mathcal{C}))$ (warning: this is *not* equivalent to $\mathsf{Add}(\mathsf{Kar}(\mathcal{C}))$).

Every linear abelian category is Cauchy complete; the converse is not true.

1.8 Simple objects (non-abelian version)

If X is an object of a $\Bbbk$-linear category, then $\mathrm{End}(X)$ is naturally a $\Bbbk$-algebra with unit the identity. It is not hard to see that the following items are equivalent:

(i) the map $\Bbbk \to \mathrm{End}_{\mathcal{C}}(X),\ k \mapsto k \cdot \mathrm{id}$ is an isomorphism of $\Bbbk$-modules;

(ii) the map $\Bbbk \to \mathrm{End}_{\mathcal{C}}(X),\ k \mapsto k \cdot \mathrm{id}$ is an isomorphism of $\Bbbk$-algebras;

(iii) the $\Bbbk$-algebra $\mathrm{End}_{\mathcal{C}}(X)$ is isomorphic to $\Bbbk$;

(iv) the $\Bbbk$-module $\mathrm{End}_{\mathcal{C}}(X)$ is free of rank 1.

If an object X satisfies any of these equivalent descriptions, we say that X is simple.

1.9 Semisimple category (non-abelian version)

Suppose $\mathcal{C}$ is a Cauchy complete linear category whose isomorphism classes of objects form a set. Then the following conditions are equivalent:

(S) There is a set S and an equivalence of categories $\mathcal{C} \cong \mathrm{Vec}_{\mathrm{fd}}(S)$ (the category of S-graded vector spaces and grading-preserving maps).

(End) For every object $X \in \mathcal{C}$, the endomorphism space $\mathrm{End}(X)$ is a finite-dimensional complex semisimple algebra, i.e., a multimatrix algebra.

(Obj) Every $X \in \mathcal{C}$ is isomorphic to a finite direct sum of simple objects $X = \bigoplus_{i=1}^{n} X_i$, where each pair X_i, X_j are either isomorphic or distinct.

(Müger) There is a set of pairwise distinct simple objects $\{X_s\}_{s\in S}$ such that for any $A, B \in \mathcal{C}$, the composition map

$$\bigoplus_{s\in S} \mathcal{C}(A \to X_s) \otimes_{\Bbbk} \mathcal{C}(X_s \to B) \xrightarrow{\circ} \mathcal{C}(A \to B)$$

is an isomorphism. (The direct sum is taken in the category of finite-dimensional vector spaces.)

Being *finitely* semisimple means that S is a finite set, i.e., there are finitely many isomorphism classes of simple objects.

Fact: A Cauchy complete linear category is semisimple if and only if it is abelian and all short exact sequences split.

More important fact: Linearity and object semisimplicity already imply abelianness! See Section 4.4.

2 Limits and Colimits

2.1 Kernel (via equalizer/pullback)

Let $\mathcal{C}$ be a category with a zero object 0 (an object that is both initial and terminal) and suppose equalizers exist in $\mathcal{C}$. For a morphism $f : X \to Y$, the zero morphism $0_{XY} : X \to Y$ is the unique factorization of $X \to 0 \to Y$.

The **kernel** of f can be defined as the equalizer of the pair of parallel morphisms f and 0_{XY}:

$$\ker(f) \xrightarrow{k} X \rightrightarrows^{f}_{0_{XY}} Y.$$

That is, $k : \ker(f) \to X$ is an object and morphism such that $f \circ k = 0_{XY} \circ k (= 0_{\ker(f),Y})$ and it is universal with this property (any other $g : Z \to X$ with $f \circ g = 0_{XY} \circ g$ factors uniquely through k).

Since 0 is initial, the morphism $0 \to X$ exists uniquely. Since 0 is terminal, the morphism $X \to 0$ exists uniquely. Their composite $X \to 0 \to Y$ is the zero morphism 0_{XY}.

Alternatively, in a category $\mathcal{C}$ with a zero object 0 (which is both initial and terminal) and pullbacks, the **kernel** of a morphism $f : X \to Y$ can be defined using a pullback.

Specifically, the kernel $k : \ker(f) \to X$ is the pullback of f along the unique zero morphism $0_Y : 0 \to Y$. The pullback diagram is:

$$\begin{tikzcd} \ker(f) \arrow[r] \arrow[d, "k"'] & 0 \arrow[d, "0_Y"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

The map k automatically satisfies $f \circ k = 0$, and the universal property of the pullback ensures that k is universal among all maps into X whose composition with f is zero.

2.2 Cokernel (via Coequalizer)

Dually, let $\mathcal{C}$ be a category with a zero object 0 and suppose coequalizers exist. For a morphism $f : X \to Y$, consider the zero morphism $0_{XY} : X \to Y$.

The **cokernel** of f can be defined as the coequalizer of the pair f and 0_{XY}:

$$X \rightrightarrows^{f}_{0_{XY}} Y \xrightarrow{q} \mathrm{coker}(f).$$

That is, $q : Y \to \mathrm{coker}(f)$ is an object and morphism such that $q \circ f = q \circ 0_{XY} (= 0_{X,\mathrm{coker}(f)})$ and it is universal with this property (any other $g : Y \to Z$ with $g \circ f = g \circ 0_{XY}$ factors uniquely through q).

2.3 Left/Right Exact Functors

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between categories where certain limits or colimits exist.

The functor F is **left exact** if it preserves all finite limits that exist in $\mathcal{C}$. This means that if L is the limit of a finite diagram $D : J \to \mathcal{C}$ (where J is a finite category), then $F(L)$ is the limit of the diagram $F \circ D : J \to \mathcal{D}$. Common examples include preserving terminal objects, products, and equalizers (and thus pullbacks and kernels if they exist).

The functor F is **right exact** if it preserves all finite colimits that exist in $\mathcal{C}$. This means that if C is the colimit of a finite diagram $D : J \to \mathcal{C}$, then $F(C)$ is the colimit of the diagram

$F \circ D : J \to \mathcal{D}$. Common examples include preserving initial objects, coproducts, and coequalizers (and thus pushouts and cokernels if they exist).

A functor is **exact** if it is both left exact and right exact (preserves all finite limits and finite colimits).

In the context of abelian categories, these definitions are equivalent to the definitions involving preservation of kernels/cokernels or exact sequences, assuming the functor is also additive.

2.4 Pullbacks from Products and Equalizers

The existence of pullbacks in a category $\mathcal{C}$ is closely related to the existence of binary products and equalizers.

Specifically, if a category $\mathcal{C}$ has all binary products and all equalizers, then it has all pullbacks. To construct the pullback of a diagram $X \xrightarrow{f} Z \xleftarrow{g} Y$:

1. Form the binary product $X \times Y$ with its projection morphisms $p_X : X \times Y \to X$ and $p_Y : X \times Y \to Y$.
2. Consider the two morphisms from the product to Z: $f \circ p_X : X \times Y \to Z$ and $g \circ p_Y : X \times Y \to Z$.
3. Take the equalizer $e : P \to X \times Y$ of this pair:

$$P \xrightarrow{e} X \times Y \rightrightarrows^{f \circ p_X}_{g \circ p_Y} Z.$$

The resulting object P together with the morphisms $p_1 = p_X \circ e : P \to X$ and $p_2 = p_Y \circ e : P \to Y$ forms the pullback square:

$$\begin{array}{ccc} P & \xrightarrow{p_2} & Y \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xrightarrow[f]{} & Z \end{array}$$

Conversely, if a category has all pullbacks and a terminal object (which acts as the empty product), it has all finite products. If it has pullbacks and products, it also has equalizers (an equalizer of $h, k : A \to B$ is the pullback of $(h, k) : A \to B \times B$ along the diagonal $\Delta_B : B \to B \times B$).

Therefore, a category has all finite limits (equivalently, all pullbacks and a terminal object) if and only if it has all binary products and all equalizers (and a terminal object).

2.5 Pushouts from Coproducts and Coequalizers

Dually, the existence of pushouts is related to the existence of binary coproducts and coequalizers.

If a category $\mathcal{C}$ has all binary coproducts and all coequalizers, then it has all pushouts. To construct the pushout of a diagram $X \xleftarrow{f} Z \xrightarrow{g} Y$:

1. Form the binary coproduct $X + Y$ (often written $X \amalg Y$) with its injection morphisms $i_X : X \to X + Y$ and $i_Y : Y \to X + Y$.
2. Consider the two morphisms from Z to the coproduct: $i_X \circ f : Z \to X + Y$ and $i_Y \circ g : Z \to X + Y$.
3. Take the coequalizer $q : X + Y \to P$ of this pair:

$$Z \rightrightarrows^{i_X \circ f}_{i_Y \circ g} X + Y \xrightarrow{q} P.$$

The resulting object P together with the morphisms $q_1 = q \circ i_X : X \to P$ and $q_2 = q \circ i_Y : Y \to P$ forms the pushout square:

$$\begin{tikzcd}
Z \arrow[r, "f"] \arrow[d, "g"'] & X \arrow[d, "q_1"] \\
Y \arrow[r, "q_2"'] & P
\end{tikzcd}$$

Conversely, if a category has all pushouts and an initial object (empty coproduct), it has all finite coproducts. If it has pushouts and coproducts, it also has coequalizers (constructed dually to equalizers using pushouts and codiagonals).

Therefore, a category has all finite colimits (equivalently, all pushouts and an initial object) if and only if it has all binary coproducts and all coequalizers (and an initial object).

2.6 Ends and coends

These concepts are fundamental tools for working with functors $T : \mathcal{C}^{op} \times \mathcal{C} \to \mathcal{D}$. Ends and coends provide universal objects capturing information "along the diagonal" $T(c, c)$, defined via dinatural transformations.

Dinatural Transformations (cf. T-V B.1.1)

Let $\mathcal{C}$ be a category and $T : \mathcal{C}^{op} \times \mathcal{C} \to \mathcal{D}$ be a functor.

- A **dinatural transformation from an object D of $\mathcal{D}$ to T** is a family of morphisms

  $$\alpha = \{\alpha_c : D \to T(c, c)\}_{c \in \mathrm{Ob}(\mathcal{C})}$$

  such that for every morphism $f : c \to d$ in $\mathcal{C}$, the following diagram commutes:

  $$\begin{tikzcd}
  D \arrow[r, "\alpha_c"] \arrow[d, "\alpha_d"'] & T(c,c) \arrow[d, "T(1_c,f)"] \\
  T(d,d) & T(d,c) \arrow[l, "T(f,1_d)"']
  \end{tikzcd}$$

  The commutativity condition is $T(1_c, f) \circ \alpha_c = T(f, 1_d) \circ \alpha_d$. This family is denoted $\alpha : D \doteqdot T$ (or $\alpha : \Delta_D \doteqdot T$).

- A **dinatural transformation from T to an object D of $\mathcal{D}$** is a family of morphisms

  $$\beta = \{\beta_c : T(c, c) \to D\}_{c \in \mathrm{Ob}(\mathcal{C})}$$

  such that for every morphism $f : c \to d$ in $\mathcal{C}$, the following diagram commutes:

  $$\begin{tikzcd}
  T(d,c) \arrow[r, "T(1_d,f)"] \arrow[d, "T(f,1_c)"'] & T(d,d) \arrow[d, "\beta_d"] \\
  T(c,c) \arrow[r, "\beta_c"'] & D
  \end{tikzcd}$$

  The commutativity condition is $\beta_d \circ T(1_d, f) = \beta_c \circ T(f, 1_c)$. This family is denoted $\beta : T \doteqdot D$ (or $\beta : T \doteqdot \Delta_D$).

(Note: The general definition of a dinatural transformation between two functors $S, T : \mathcal{C}^{op} \times \mathcal{C} \to \mathcal{D}$ requires a hexagonal diagram, as shown in T-V B.1.1. The square diagrams above are special cases for constant source/target functors.)

Coends (T-V B.1.3)

Let $T : \mathcal{C}^{op} \times \mathcal{C} \to \mathcal{D}$ be a functor. The **coend** of T, denoted $\int^c T(c,c)$, is an object of $\mathcal{D}$ equipped with a dinatural transformation

$$\omega' : T \ddot{-} \int^c T(c,c)$$

satisfying the universal property:

For any object $x \in \mathcal{D}$ and any dinatural transformation $\beta : T \ddot{-} x$, there exists a **unique** morphism

$$h : \int^c T(c,c) \to x$$

such that $\beta_c = h \circ \omega'_c$ for all c:

$$\begin{tikzcd} T(c,c) \arrow[r, "\omega'_c"] \arrow[d, "\beta_c"'] & \int^c T(c,c) \arrow[d, "h"] \\ x \arrow[r, equal] & x \end{tikzcd}$$

The coend is the universal recipient for dinatural maps *from* T. If $\mathcal{D}$ has the necessary colimits, it can be constructed as a coequalizer:

$$\int^c T(c,c) = \operatorname{coeq}\left(\coprod_{f:c\to d} T(c,d) \rightrightarrows \coprod_c T(c,c) \right)$$

where the parallel arrows are induced by the morphisms $T(1_c, f)$ and $T(f, 1_d)$ into the respective summands.

- **Example (T-V B.1.3, Ex 1):** Tensor product

  $$F \otimes_{\mathcal{C}} G = \int^c (F(c) \times G(c))$$

  for $F : \mathcal{C}^{op} \to \mathbf{Set}$ and $G : \mathcal{C} \to \mathbf{Set}$.

Ends (T-V B.1.2)

Let $T : \mathcal{C}^{op} \times \mathcal{C} \to \mathcal{D}$ be a functor. The **end** of T, denoted $\int_c T(c,c)$, is an object of $\mathcal{D}$ equipped with a dinatural transformation

$$\omega : \int_c T(c,c) \ddot{-} T$$

satisfying the universal property:

For any object $x \in \mathcal{D}$ and any dinatural transformation $\alpha : x \ddot{-} T$, there exists a **unique** morphism

$$h : x \to \int_c T(c,c)$$

such that $\alpha_c = \omega_c \circ h$ for all c:

$$\begin{tikzcd} x \arrow[r, "h"] \arrow[d, equal] & \int_c T(c,c) \arrow[d, "\omega_c"] \\ x \arrow[r, "\alpha_c"'] & T(c,c) \end{tikzcd}$$

The end is the universal source for dinatural maps *to* T. If $\mathcal{D}$ has the necessary limits, it can be constructed as an equalizer:

$$\int_c T(c,c) = \mathrm{eq}\left(\prod_c T(c,c) \rightrightarrows \prod_{f:c\to d} T(d,c)\right)$$

where the parallel arrows arise from the dinaturality condition applied to the projections $\pi_c : \prod_b T(b,b) \to T(c,c)$. The first map's f-component is $T(f, 1_c) \circ \pi_c$, and the second map's f-component is $T(1_d, f) \circ \pi_d$.

- **Example (T-V B.1.2, Ex 1):** Natural transformations

$$\mathrm{Nat}(F,G) = \int_c \mathrm{Hom}_{\mathcal{D}}(Fc, Gc).$$

Properties (cf. Turaev–Virelizier Appendix B)

- **Hom-Set Isomorphisms (T-V B.1.4):** Mapping into an end or out of a coend corresponds to taking the end of the relevant Hom-functor in **Set**:

$$\mathrm{Hom}_{\mathcal{D}}\big(x, \int_c T(c,c)\big) \cong \int_c \mathrm{Hom}_{\mathcal{D}}(x, T(c,c))$$

$$\mathrm{Hom}_{\mathcal{D}}\big(\int^c T(c,c), x\big) \cong \int_c \mathrm{Hom}_{\mathcal{D}}(T(c,c), x).$$

- **Fubini Theorem (T-V B.1.5):** Under suitable conditions, the order of integration can be swapped:

$$\int_{c\in\mathcal{C}} \int_{d\in\mathcal{D}} T(c,d,c,d) \cong \int_{d\in\mathcal{D}} \int_{c\in\mathcal{C}} T(c,d,c,d),$$

$$\int^{c\in\mathcal{C}} \int^{d\in\mathcal{D}} T(c,d,c,d) \cong \int^{d\in\mathcal{D}} \int^{c\in\mathcal{C}} T(c,d,c,d).$$

- **Other properties** include preservation by certain functors and relations under adjunctions (T-V B.1.6).

3 Monoidal Categories and Related Structures

3.1 Monoidal category

A tuple $(\mathcal{C}, \otimes, \mathbf{1}, a, l, r)$ of a category $\mathcal{C}$ with a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$, an object $\mathbf{1}$ and natural isomorphisms

$$a_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z), \quad l_X : \mathbf{1} \otimes X \to X, \quad r_X : X \otimes \mathbf{1} \to X$$

satisfying the so-called pentagon equation and the triangle equation.

3.2 Left rigid category

A monoidal category in which every object admits a left dual, that is, for every object X there exists a pair $({}^\vee X, \mathrm{ev}_X)$ where ${}^\vee X$ is another object and a left evaluation

$$\mathrm{ev}_X : {}^\vee X \otimes X \to \mathbf{1}$$

is a non-degenerate pairing. This means that there exists another map

$$\mathrm{coev}_X : \mathbf{1} \to X \otimes {}^\vee X$$

called the left coevaluation satisfying

$$(\mathrm{Id} \otimes \mathrm{ev}_X)(\mathrm{coev}_X \otimes \mathrm{Id}) = \mathrm{Id}, \quad (\mathrm{ev}_X \otimes \mathrm{Id})(\mathrm{Id} \otimes \mathrm{coev}_X) = \mathrm{Id}.$$

3.3 Right rigid category

A monoidal category in which every object admits a right dual, that is, for every object X there exists a pair $(X^\vee, \widetilde{\mathrm{ev}}_X)$ where $X^\vee$ is another object and a right evaluation

$$\widetilde{\mathrm{ev}}_X : X \otimes X^\vee \to \mathbf{1}$$

is a non-degenerate pairing. This means that there exists another map

$$\widetilde{\mathrm{coev}}_X : \mathbf{1} \to X^\vee \otimes X$$

called the right coevaluation satisfying

$$(\mathrm{Id} \otimes \widetilde{\mathrm{ev}}_X)(\widetilde{\mathrm{coev}}_X \otimes \mathrm{Id}) = \mathrm{Id}, \quad (\widetilde{\mathrm{ev}}_X \otimes \mathrm{Id})(\mathrm{Id} \otimes \widetilde{\mathrm{coev}}_X) = \mathrm{Id}.$$

3.4 Rigid or autonomous category

A monoidal category which is both left and right rigid.

3.5 Pivotal or sovereign category

A rigid category in which the left and right dual functors coincide as monoidal functors. In such a case we put $X^* =^\vee X = X^\vee$.

Alternatively, it can be defined as a left rigid category endowed with a distinguished monoidal natural isomorphism

$$X \to {}^{\vee\vee}X.$$

3.6 Spherical category (classical definition)

If $\mathcal{C}$ is a pivotal category, we have the notions of left and right traces of an endomorphism $f : X \to X$, defined as follows:

$$\mathrm{tr}_l(f) := \mathrm{ev}_X(\mathrm{Id}_{X^*} \otimes f)\widetilde{\mathrm{coev}}_X : \mathbf{1} \to \mathbf{1}, \quad \mathrm{tr}_r(f) := \widetilde{\mathrm{ev}}_X(f \otimes \mathrm{Id}_{X^*})\mathrm{coev}_X : \mathbf{1} \to \mathbf{1}.$$

We also have the notions of left and right dimensions of an object X:

$$\dim_l(X) := \mathrm{ev}_X\widetilde{\mathrm{coev}}_X, \quad \dim_r(X) := \widetilde{\mathrm{ev}}_X\mathrm{coev}_X.$$

A spherical category is a pivotal category whose left and right traces coincide (and therefore also the left and right dimensions).

3.7 Braided category

A monoidal category endowed with a natural isomorphism

$$\tau_{X,Y} : X \otimes Y \to Y \otimes X$$

which is $\otimes$-multiplicative in the sense that

$$\tau_{X,Y\otimes Z} = (\mathrm{id}_Y \otimes \tau_{X,Z})(\tau_{X,Y} \otimes \mathrm{id}_Z), \quad \tau_{X\otimes Y,Z} = (\tau_{X,Z} \otimes \mathrm{id}_Y)(\mathrm{id}_X \otimes \tau_{Y,Z})$$

3.8 Ribbon or tortile category (take 1)

Let $\mathcal{C}$ be a braided pivotal category. The left twist is defined as the following family of morphisms (in fact natural isomorphisms):

$$\theta^l_X := (\mathrm{ev}_X \otimes \mathrm{id}_X)(\mathrm{id}_{X^*} \otimes \tau_{X,X})(\widetilde{\mathrm{coev}_X} \otimes \mathrm{id}_X) : X \to X$$

whereas the right twist is

$$\theta^r_X := (\mathrm{id}_X \otimes \widetilde{\mathrm{ev}}_X)(\tau_{X,X} \otimes \mathrm{id}_{X^*})(\mathrm{id}_X \otimes \mathrm{coev}_X) : X \to X.$$

A ribbon category is a braided pivotal category in which the left and right twists coincide, $\theta = \theta^l = \theta^r$. It automatically satisfies that the twist is self-dual in the sense that $(\theta_X)^* = \theta_{X^*}$.

3.9 Balanced category

A braided category together with a twist, that is, a natural isomorphism $\theta : X \to X$, called the twist, satisfying

$$\theta_{X\otimes Y} = \tau_{Y,X}\tau_{X,Y}(\theta_X \otimes \theta_Y).$$

Key property: Braided pivotal = balanced rigid.

3.10 Ribbon or tortile category (take 2)

A balanced left rigid category in which the twist and the rigid structure are compatible in the sense that

$$^\vee(\theta_X) = \theta_{^\vee X}.$$

3.11 Monoidal $\Bbbk$-linear category

A monoidal category which is also $\Bbbk$-linear and whose monoidal product is $\Bbbk$-bilinear.

3.12 Pre-fusion category (non-abelian version)

A pre-fusion $\Bbbk$-category is a monoidal $\Bbbk$-linear category $\mathcal{C}$ with the property that there exists a collection of simple objects $(V_i)_{i\in I}$ indexed by some set I satisfying that:

(a) the monoidal unit $\mathbf{1}$ belongs to the family (and it is typically indexed as $V_0 = \mathbf{1}$);

(b) $\mathrm{Hom}_{\mathcal{C}}(V_i, V_j) = 0$ if $i \neq j$;

(c) every object of $\mathcal{C}$ is a direct sum of finitely many objects from the indexed collection $(V_i)_{i\in I}$.

It readily follows that every simple object in $\mathcal{C}$ is isomorphic to exactly one object from $(V_i)_{i\in I}$. Also, (b) and (c) imply that the hom-sets are finite-free $\Bbbk$-modules and that for any pair of objects X, Y:

$$\mathrm{Hom}_{\mathcal{C}}(X, Y) \cong \bigoplus_{i\in I} \mathrm{Hom}_{\mathcal{C}}(X, V_i) \otimes_{\Bbbk} \mathrm{Hom}_{\mathcal{C}}(V_i, Y).$$

3.13 Fusion category (non-abelian version)

A pre-fusion rigid linear category in which the index set I is finite.

The dimension of a pivotal fusion linear category is given by:

$$\dim(\mathcal{C}) := \sum_{i \in I} \dim_l(V_i) \dim_r(V_i),$$

and if the category happens to be spherical then this formula becomes:

$$\dim(\mathcal{C}) = \sum_{i \in I} (\dim_l(V_i))^2.$$

3.14 Modular category (non-abelian, fusion version)

A ribbon fusion $\Bbbk$-linear category with the property that, if:

$$s_{i,j} := \mathrm{tr}(\tau_{V_j,V_i} \tau_{V_i,V_j}) \in \mathrm{End}_{\mathcal{C}}(\mathbf{1}) = \Bbbk$$

(where τ denotes the braiding), then the S-matrix $S = (s_{ij})_{i,j \in I}$ is invertible over $\Bbbk$.

3.15 Drinfeld center

Let $(\mathcal{C}, \otimes, 1)$ be a monoidal category. The **Drinfeld center** $\mathcal{Z}(\mathcal{C})$ is defined as follows:

Objects: Pairs (Z, β_Z), where $Z \in \mathcal{C}$ and $\beta_Z = \{\beta_{Z,X} : Z \otimes X \xrightarrow{\cong} X \otimes Z\}_{X \in \mathrm{Ob}(\mathcal{C})}$ is a family of natural isomorphisms (called the *half-braiding*) satisfying:

- Naturality: for any $f : X \to Y$, the diagram

$$\begin{tikzcd}
Z \otimes X \arrow[r, "\beta_{Z,X}"] \arrow[d, "\mathrm{id}_Z \otimes f"'] & X \otimes Z \arrow[d, "f \otimes \mathrm{id}_Z"] \\
Z \otimes Y \arrow[r, "\beta_{Z,Y}"] & Y \otimes Z
\end{tikzcd}$$

  commutes.

- Coherence: $\beta_{Z,X \otimes Y} = (\mathrm{id}_X \otimes \beta_{Z,Y}) \circ (\beta_{Z,X} \otimes \mathrm{id}_Y)$;

- Unit compatibility: $\beta_{Z,1} = \mathrm{id}_Z$.

Morphisms: A morphism $f : (Z, \beta_Z) \to (W, \beta_W)$ is a morphism $f : Z \to W$ in $\mathcal{C}$ such that:

$$(\mathrm{id}_X \otimes f) \circ \beta_{Z,X} = \beta_{W,X} \circ (f \otimes \mathrm{id}_X)$$

for all $X \in \mathcal{C}$, i.e.,

$$\begin{tikzcd}
Z \otimes X \arrow[r, "\beta_{Z,X}"] \arrow[d, "f \otimes \mathrm{id}_X"'] & X \otimes Z \arrow[d, "\mathrm{id}_X \otimes f"] \\
W \otimes X \arrow[r, "\beta_{W,X}"] & X \otimes W
\end{tikzcd}$$

Monoidal structure: $\mathcal{Z}(\mathcal{C})$ is a braided monoidal category with:

- Tensor product: $(Z, \beta_Z) \otimes (W, \beta_W) := (Z \otimes W, \beta_{Z\otimes W})$, where:

$$\beta_{Z\otimes W,X} = (\beta_{Z,X} \otimes \mathrm{id}_W) \circ (\mathrm{id}_Z \otimes \beta_{W,X})$$

- Unit: $(1, \beta_1)$, where $\beta_{1,X} = \mathrm{id}_X$;

- Braiding: $c_{(Z,\beta_Z),(W,\beta_W)} = \beta_{Z,W}$.

3.16 Müger center

Let $(\mathcal{C}, \otimes, 1, c)$ be a braided monoidal category. The **Müger center** $\mathcal{Z}_2(\mathcal{C})$ is the full subcategory of $\mathcal{C}$ whose objects Z satisfy:

$$c_{X,Z} \circ c_{Z,X} = \mathrm{id}_{Z\otimes X} \quad \text{for all } X \in \mathcal{C}.$$

$\mathcal{Z}_2(\mathcal{C})$ inherits the monoidal structure and is a symmetric monoidal category.

3.17 Equivalent Characterizations of Modular Categories

Theorem: Let $\mathcal{C}$ be a braided fusion category over an algebraically closed field $\Bbbk$. Let $\{X_i\}_{i\in I}$ be a complete set of representatives of isomorphism classes of simple objects with $X_0 = \mathbf{1}$. Let $c_{X,Y}$ denote the braiding and $\overline{\mathcal{C}}$ the same category with inverse braiding.

The following are equivalent:

(i) **Trivial Müger center:** $\mathcal{Z}_2(\mathcal{C}) \cong \mathrm{Vec}_\Bbbk$;

(ii) **Invertible S-matrix:** The matrix $S = (S_{ij})$, with entries

$$S_{ij} = \mathrm{Tr}(c_{X_j,X_i} \circ c_{X_i,X_j})$$

is invertible over $\Bbbk$;

(iii) **Drinfeld center equivalence:** $\mathcal{Z}(\mathcal{C}) \cong \mathcal{C} \boxtimes \overline{\mathcal{C}}$.

Remarks:

- The pair (S, T), with $T_{ii} = \theta_i$, is called the *modular data*.

- These equivalences are due to Müger.

- Condition (i) expresses maximal non-degeneracy.

- Condition (iii) shows that the center is determined by the braiding and its inverse.

4 Abelian Categories and Related Structures

4.1 Kernels and cokernels

Let $\mathcal{C}$ be a category with a zero object, and let $f : X \to Y$ be a morphism. A *kernel* of f is a pair $(\ker(f), i : \ker(f) \to X)$ such that $f \circ i = 0$, and it is universal with respect to this property: if $g : Z \to X$ satisfies $f \circ g = 0$, then there exists a unique $g' : Z \to \ker(f)$ such that $i \circ g' = g$:

$$\begin{tikzcd} & Z \arrow[dl, dashed, "\exists!\ g'"'] \arrow[d, "g"] & \\ \ker(f) \arrow[r, "i"] & X \arrow[r, "f"] & Y \end{tikzcd}$$

A *cokernel* of f is a pair $(\mathrm{coker}(f), \pi : Y \to \mathrm{coker}(f))$ such that $\pi \circ f = 0$, and it is universal with respect to this property: if $g : Y \to Z$ satisfies $g \circ f = 0$, then there exists a unique $g' : \mathrm{coker}(f) \to Z$ such that $g' \circ \pi = g$:

$$\begin{tikzcd} X \arrow[r, "f"] & Y \arrow[r, "\pi"] \arrow[d, "g"'] & \mathrm{coker}(f) \arrow[dl, dashed, "\exists!\ g'"] \\ & Z & \end{tikzcd}$$

A kernel of a cokernel is called the *image* of f, denoted $\mathrm{im}(f) \to Y$. A cokernel of a kernel is called the *coimage*, denoted $X \to \mathrm{coim}(f)$. The universal properties guarantee uniqueness up to unique isomorphism.

4.2 Monos and epis

If $\mathcal{C}$ is an arbitrary category, a morphism $i : X \to Y$ is a *monomorphism* (or *monic*) if $i \circ f = i \circ g$ for morphisms $f, g : Z \to X$ implies $f = g$. Dually, a morphism $\pi : X \to Y$ is an *epimorphism* (or *epi*) if $f \circ \pi = g \circ \pi$ for morphisms $f, g : Y \to Z$ implies $f = g$.

If $\mathcal{C}$ is additive, then:

- $i : X \to Y$ is monic if and only if $i \circ f = 0$ implies $f = 0$;
- $\pi : X \to Y$ is epi if and only if $f \circ \pi = 0$ implies $f = 0$.

Another characterisation (still assuming additivity):

- $i : X \to Y$ is monic if and only if the unique morphism $0 \to X$ is a kernel of i;
- $\pi : X \to Y$ is epi if and only if the unique morphism $Y \to 0$ is a cokernel of π.

4.3 Abelian category

If $\mathcal{C}$ is an additive category, one can easily see that all kernels are monic and all cokernels are epi. The converse might not hold. When it does, then we talk of an abelian category.

More precisely, an additive category $\mathcal{C}$ is *abelian* if:

(a) every morphism has a kernel and a cokernel,

(b) every monomorphism is the kernel of its cokernel,

(c) every epimorphism is the cokernel of its kernel.

If $F : \mathcal{A} \to \mathcal{B}$ is a functor between abelian categories, then:

- F is *left exact* if it is additive and preserves kernels: $F(\ker(f)) \cong \ker(Ff)$;
- F is *right exact* if it is additive and preserves cokernels: $F(\mathrm{coker}(f)) \cong \mathrm{coker}(Ff)$;
- F is *exact* if it is both left and right exact.

Exactness can also be expressed in terms of short exact sequences, where these are defined in abelian categories using kernels and images.

4.4 Linear + object semisimple imply abelian

This proof uses the key consequence that in a $\Bbbk$-linear semisimple category, every monomorphism and every epimorphism splits.

Assumptions

- $\mathcal{C}$ is $\Bbbk$-linear with finite biproducts ($\oplus$).
- $\mathcal{C}$ is semisimple (every object is finite $\oplus$ of simples S_i, Schur's Lemma holds).
- **Key Consequence:** Every mono splits; every epi splits.

Goal: Show $\mathcal{C}$ is Abelian

(A) Zero Object: Exists (empty $\oplus$).

(B) Finite Biproducts: Exists (by assumption).

(C) Existence of Kernels and Cokernels:

- **Kernel of $f : X \to Y$:**
  1. Decompose $X \cong \bigoplus_{i \in I} S_i$. Let $J = \{i \in I \mid f \circ \mathrm{incl}_{S_i} = 0\}$.
  2. Define $K = \bigoplus_{j \in J} S_j$ and let $k : K \to X$ be the canonical inclusion.
  3. Check $f \circ k = 0$: True by definition of J.
  4. Check Universality: Let $g : Z \to X$ satisfy $f \circ g = 0$. Since $k : K \to X$ is a monomorphism (inclusion of summands), it splits by the Key Consequence. Let $p_K : X \to K$ be a projection such that $p_K \circ k = \mathrm{id}_K$. Define $h = p_K \circ g : Z \to K$. We must show $k \circ h = g$. Let $X \cong K \oplus K'$ where $k' : K' \to X$. Any map $g : Z \to X$ corresponds to a pair $(g_K, g_{K'}) = (p_K \circ g, p_{K'} \circ g)$.
     We have
     $$f \circ g = f \circ (k \circ g_K + k' \circ g_{K'}) = (f \circ k) \circ g_K + (f \circ k') \circ g_{K'} = 0 + (f \circ k') \circ g_{K'} = 0.$$
     By construction of K, f restricted to K' (via $f \circ k'$) is non-zero on the simple components of K'. This forces $g_{K'} = 0$.
     So g corresponds to $(g_K, 0)$, which means $g = k \circ g_K$. Taking $h = g_K = p_K \circ g$ gives $k \circ h = g$. Uniqueness of h follows because k is mono.

5. Kernels exist.

- **Cokernel of $f : X \to Y$:**
  1. Let $I \subseteq Y$ be the subobject generated by the images of all simple summands of X under f. (Technically, $I = \sum \mathrm{Im}(f \circ \mathrm{incl}_{S_i})$.) Since $\mathcal{C}$ is semisimple, the subobject I is a direct summand: $Y \cong I \oplus C$.
  2. Let $p : Y \to C$ be the projection onto C.
  3. Check $p \circ f = 0$: The image of f lies entirely within I. The projection p annihilates I. So $p \circ f = 0$.
  4. Check Universality: Let $h : Y \to Z$ satisfy $h \circ f = 0$. This implies h restricted to I is zero ($h \circ \mathrm{incl}_I = 0$). Since $p : Y \to C$ is an epimorphism (projection from a direct sum), it splits by the Key Consequence. Let $s : C \to Y$ be a section ($p \circ s = \mathrm{id}_C$). Define $q = h \circ s : C \to Z$. We must show $q \circ p = h$.

     Any map $h : Y \to Z$ corresponds to a pair (h_I, h_C) where $h_I = h \circ \mathrm{incl}_I$ and $h_C = h \circ s$. The condition $h \circ f = 0$ implies $h_I = 0$. So h corresponds to $(0, h_C)$.

     $q = h \circ s = h_C$.

     $q \circ p$ takes $y = (y_I, y_C)$, maps via p to y_C, maps via q to $h_C(y_C)$. This corresponds to the map $(0, h_C)$, which is h. Yes. Uniqueness of q follows because p is epi.
  5. Cokernels exist.

(D) Monos are Kernels, Epis are Cokernels:

- Let $m : A \to X$ be mono. It splits, so $X \cong A \oplus C$. Let $p : X \to C$ be the projection. By the kernel construction in (C), $K = \ker(p)$ is the sum of simple summands of X annihilated by p, which is exactly A. So
$$m = \ker(p) = \ker(\mathrm{coker}(m)).$$
- Let $e : X \to C$ be epi. It splits, so $X \cong K \oplus C$. Let $k : K \to X$ be inclusion. By the cokernel construction in (C), $P = \mathrm{coker}(k)$ is the projection onto the summand complementary to K, which is C. So
$$e = \mathrm{coker}(k) = \mathrm{coker}(\ker(e)).$$

Conclusion: Since axioms (A)–(D) hold, $\mathcal{C}$ is an abelian category.

4.5 Monoidal abelian category

A category which is monoidal and abelian with the following compatibility conditions: the tensor product $\otimes$ is a biadditive functor (i.e., additive in each variable).

4.6 Monoidal abelian $\Bbbk$-linear category

A category which is monoidal, abelian and $\Bbbk$-linear with the following compatibility conditions: the $\Bbbk$-module enrichment extends the abelian group enrichment, the tensor product $\otimes$ is a $\Bbbk$-bilinear functor, and the unit $\mathbf{1}$ is a simple object with $\mathrm{End}(\mathbf{1}) = \Bbbk$.

4.7 Projective and injective objects

An object A in an abelian category $\mathcal{A}$ is *projective* if for every epimorphism $p : X \to Y$ and every morphism $f : A \to Y$, there is a morphism $f' : A \to X$ that lifts p, i.e., $f = p \circ f'$:

$$\begin{tikzcd} & X \arrow[d, "p"] \\ A \arrow[ur, dashed, "\exists f'"] \arrow[r, "f"'] & Y \end{tikzcd}$$

An object A in an abelian category $\mathcal{A}$ is *injective* if for every monomorphism $i : Y \to X$ and every morphism $f : Y \to A$, there is a morphism $f' : X \to A$ that extends i, i.e., $f = f' \circ i$:

$$\begin{tikzcd} Y \arrow[r, "f"] \arrow[d, "i"'] & A \\ X \arrow[ur, dashed, "\exists f'"'] & \end{tikzcd}$$

One can show, as for the category of modules, that an object A is projective if and only if the functor

$$\mathrm{Hom}_{\mathcal{A}}(A, -) : \mathcal{A} \to \mathsf{Ab}$$

is exact. Similarly, an object A is injective if and only if the functor

$$\mathrm{Hom}_{\mathcal{A}}(-, A) : \mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$$

is exact.

In the abelian category of R-modules, free $\Rightarrow$ projective.

In the abelian category of $\mathbb{Z}$-modules (i.e., abelian groups), divisible $\Rightarrow$ injective (recall that an abelian group A is divisible if for every non-zero integer n, the canonical map $A \to A$, $a \mapsto na$ is surjective. If A is torsion-free, then this is the same as A being uniquely divisible, i.e., that map being an isomorphism, because being torsion-free means precisely that such a map is always injective.) So examples are $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$, etc.

4.8 Semisimple category

This is the analogue of pre-fusion categories for abelian categories (instead of for linear categories).

Let $\mathcal{A}$ be an abelian category and let Y be an object. A *subobject* of Y is a monomorphism $i : X \to Y$. A *quotient object* of Y is an epimorphism $\pi : Y \to Z$.

A non-zero object X is said to be

(i) *simple* if the source objects of its subobjects are all isomorphic to X or 0,

(ii) *semisimple* if it is a coproduct of simple objects,

(iii) *indecomposable* if it is not a coproduct of at least two non-zero subobjects.

An abelian category is *semisimple* if all its objects are semisimple.

Note: there are many other versions of what a semisimple category should be in other contexts different than abelian categories, see here for an overview and comparison.

4.9 Semisimplicity vs. Projectivity in Abelian Categories

Theorem: Let $\mathcal{C}$ be an abelian category. The following conditions are equivalent:

1. $\mathcal{C}$ is **semisimple** (every object is a direct sum of simple objects).
2. Every short exact sequence in $\mathcal{C}$ **splits**.
3. Every object in $\mathcal{C}$ is **projective**.
4. Every object in $\mathcal{C}$ is **injective**.
5. Every subobject of any object in $\mathcal{C}$ is a **direct summand**.

Proof Sketch for (1) $\iff$ (3)

(3) $\implies$ (1): Assume every object is projective. Show $\mathcal{C}$ is semisimple.

- It suffices to show that every short exact sequence (SES)
$$0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$$
splits.
- By assumption, the object C is projective.
- The definition of projective means that for the epimorphism $p : B \to C$ and the identity morphism $\mathrm{id}_C : C \to C$, there exists a lifting (a section) $s : C \to B$ such that $p \circ s = \mathrm{id}_C$.
- The existence of such a section s means the SES splits.
- Since every SES splits, $\mathcal{C}$ is semisimple.

(1) $\implies$ (3): Assume $\mathcal{C}$ is semisimple. Show every object P is projective.

- We need to show that for any epimorphism $p : B \to C$ and any morphism $f : P \to C$, there exists $\tilde{f} : P \to B$ such that $p \circ \tilde{f} = f$.
- Since $\mathcal{C}$ is semisimple, every SES splits. Consider the SES
$$0 \to K \xrightarrow{i} B \xrightarrow{p} C \to 0,$$
where $K = \ker(p)$. This sequence splits.
- Splitting implies $B \cong K \oplus C$. Under this isomorphism, let p correspond to the projection
$$p_C : K \oplus C \to C.$$
- Given $f : P \to C$, define $\tilde{f}' : P \to K \oplus C$ by $\tilde{f}' = (0, f)$, where $0 : P \to K$ is the zero morphism.
- Let $\tilde{f} : P \to B$ correspond to $\tilde{f}'$ via the isomorphism $B \cong K \oplus C$.
- Then
$$p \circ \tilde{f} = p_C \circ \tilde{f}' = p_C \circ (0, f) = f.$$
- We have found the required lifting $\tilde{f}$, so P is projective.

Thus, an abelian category is semisimple if and only if all of its objects are projective.

4.10 Locally finite abelian $\Bbbk$-linear category

An object A of an abelian category has *finite length* if there is a Jordan-Hölder series of finite length

$$0 = A_0 \to A_1 \to \cdots \to A_{n-1} \to A_n = A$$

of monomorphisms such that each object

$$A_{i+1}/A_i := \operatorname{coker}(A_i \to A_{i+1})$$

is simple.

If S is a simple object, we say that a given Jordan-Hölder series contains S with multiplicity m if the number of values of i for which A_{i+1}/A_i is isomorphic to S is m. The Jordan-Hölder theorem states that all Jordan-Hölder series of a given object have the same length, and more particularly, that any two Jordan-Hölder series contain any simple object with the same multiplicity. We denote $[A : S] := m$.

Let $\Bbbk$ be a field. An abelian $\Bbbk$-linear category $\mathcal{A}$ is *locally finite* if:

1. all $\mathrm{Hom}_{\mathcal{A}}(X, Y)$ are finite-dimensional $\Bbbk$-vector spaces,
2. every object has finite length.

4.11 Finite abelian $\Bbbk$-linear category

Let $\Bbbk$ be a field. A locally finite abelian $\Bbbk$-linear category $\mathcal{A}$ is *finite* if:

1. there are finitely many isomorphism classes of simple objects,
2. every simple object has a projective cover, that is, for every simple object A there exists a projective object P_A and an epimorphism $\pi : P_A \to A$ which is universal: any other epimorphism $p : P \to A$ from a projective object P factors through π, i.e., there exists an epimorphism $p' : P \to P_A$ such that $p = \pi \circ p'$.

It is a theorem by Deligne that any finite abelian $\Bbbk$-linear category is equivalent to the category Mod_A^{fd} of finite-dimensional A-modules for some finite-dimensional algebra A. More precisely, the given finite abelian $\Bbbk$-linear category determines the Morita equivalence class of A (recall that two algebras are Morita equivalent if their categories of modules are equivalent).

4.12 Left/right exactness, adjointness and (dual) representability

Let $F, G : \mathcal{C} \to \mathcal{D}$ be linear functors between finite linear categories. The following statements are equivalent:

(L1) F is left-exact.

(L2) F admits a left adjoint.

Likewise, the following are equivalent:

(R1) G is right-exact.

(R2) G admits a right adjoint.

Moreover, for $\mathcal{D} = \mathbf{vect}$, in addition the following statements are equivalent to (L1) and (L2) and to (R1) and (R2), respectively:

(L3) F is representable, i.e. $F \cong \mathrm{Hom}_{\mathcal{C}}(c, -)$ for some $c \in \mathcal{C}$.

(R3) G is 'dually representable', i.e. $G \cong \mathrm{Hom}_{\mathcal{C}}(-, d)^*$ for some $d \in \mathcal{C}$.

4.13 Deligne product

Given $\Bbbk$-linear abelian categories $\mathcal{C}, \mathcal{D}$, their Deligne product (if it exists) is another $\Bbbk$-linear abelian category $\mathcal{C} \bullet \mathcal{D}$ together with a bilinear right exact functor

$$\mathcal{C} \times \mathcal{D} \to \mathcal{C} \bullet \mathcal{D}$$

that induces equivalences

$$\mathrm{Rex}[\mathcal{C} \bullet \mathcal{D}, \mathcal{E}] \simeq \mathrm{Rex}[\mathcal{C}, \mathcal{D}; \mathcal{E}]$$

for all abelian $\mathcal{E}$. The Deligne product might not always exist.

4.14 Kelly product

Given $\Bbbk$-linear finitely cocomplete categories $\mathcal{C}, \mathcal{D}$, their Kelly product (if it exists) is another $\Bbbk$-linear finitely cocomplete category $\mathcal{C} \boxtimes \mathcal{D}$ together with a bilinear right exact functor (i.e., finite colimit preserving)

$$\mathcal{C} \times \mathcal{D} \to \mathcal{C} \boxtimes \mathcal{D}$$

that induces equivalences

$$\mathrm{Rex}[\mathcal{C} \boxtimes \mathcal{D}, \mathcal{E}] \simeq \mathrm{Rex}[\mathcal{C}, \mathcal{D}; \mathcal{E}]$$

for all $\mathcal{E}$ finitely cocomplete. It always exists. One can realise

$$\mathcal{C} \boxtimes \mathcal{D} \subset \mathrm{Lex}[\mathcal{C}^{\mathrm{op}}, \mathcal{D}^{\mathrm{op}}; \Bbbk\text{-}\mathsf{Mod}].$$

Example: If R, S are $\Bbbk$-algebras, the functor

$$\otimes_{\Bbbk} : R\text{-}\mathsf{Mod}_f \times S\text{-}\mathsf{Mod}_f \to (R \otimes_{\Bbbk} S)\text{-}\mathsf{Mod}_f$$

induces an equivalence

$$(R \otimes_{\Bbbk} S)\text{-}\mathsf{Mod}_f \simeq R\text{-}\mathsf{Mod}_f \boxtimes S\text{-}\mathsf{Mod}_f.$$

Facts: (Franco, EGNO)

(1) If $\mathcal{C}, \mathcal{D}$ are abelian, then $\mathcal{C} \boxtimes \mathcal{D}$ is abelian if and only if $\mathcal{C} \bullet \mathcal{D}$ exists, and in that case:

$$\mathcal{C} \boxtimes \mathcal{D} \simeq \mathcal{C} \bullet \mathcal{D}.$$

(2) If $\mathcal{C}, \mathcal{D}$ are locally finite abelian $\Bbbk$-linear categories, then so is $\mathcal{C} \boxtimes \mathcal{D}$.

Moral: Among locally finite abelian $\Bbbk$-linear categories, the Kelly and Deligne products exist, and they coincide. This is called the *Deligne-Kelly tensor product.*

Further properties:

(1) $\mathrm{Hom}_{\mathcal{C} \boxtimes \mathcal{D}}(X \boxtimes X', Y \boxtimes Y') \cong \mathrm{Hom}_{\mathcal{C}}(X, Y) \otimes_{\Bbbk} \mathrm{Hom}_{\mathcal{D}}(X', Y')$;

(2) A bilinear biexact functor $\mathcal{C} \times \mathcal{D} \to \mathcal{E}$ induces an exact functor $\mathcal{C} \boxtimes \mathcal{D} \to \mathcal{E}$;

(3) Right exact functors $F_i : \mathcal{C}_i \to \mathcal{D}_i$ between locally finite abelian linear categories give rise to a right exact functor:

$$F_1 \boxtimes F_2 : \mathcal{C}_1 \boxtimes \mathcal{C}_2 \to \mathcal{D}_1 \boxtimes \mathcal{D}_2.$$

4.15 (Multi)tensor, fusion and ring categories

Let $\mathcal{C}$ be a locally finite abelian $\Bbbk$-linear monoidal category.

- $\mathcal{C}$ is a **multiring** category if $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ is bilinear and biexact.
- If $\mathrm{End}(1) \cong \Bbbk$, it is a **ring** category.

If $\mathcal{C}$ is in addition rigid:

- **multitensor** if $\otimes$ is bilinear on morphisms,
- **multifusion** if it is finite semisimple multitensor,
- **tensor** if multitensor and $\mathrm{End}(1) \cong \Bbbk$,
- **fusion** if multifusion and $\mathrm{End}(1) \cong \Bbbk$.

Properties:

1. In a multiring category with left (resp. right) duals, the corresponding dualisation functor is exact.
2. A finite ring category with left duals is a tensor category.
3. If P is projective in a multiring category and X has a left (resp. right) dual, then $P \otimes X$ (resp. $X \otimes P$) is projective.
4. In a multiring category with left duals (e.g., multitensor),
$$1 \in \mathcal{C} \text{ is projective iff } \mathcal{C} \text{ is semisimple.}$$
5. In a ring category with left duals (e.g., tensor), $1 \in \mathcal{C}$ is simple; in general in multiring with left duals it is semisimple.
6. The Deligne-Kelly product of (multi)tensor, (multi)fusion, or (multi)ring categories is of such type.

4.16 The distinguished invertible object (take 1)

Let $\mathcal{C}$ be a rigid monoidal category. An object $X \in \mathcal{C}$ is **invertible** if the evaluation and coevaluation maps give isomorphisms
$$X^* \otimes X \cong 1 \quad \text{and} \quad 1 \cong X \otimes X^*.$$

Example: In the category $\mathrm{Rep}(G)$, the 1-dimensional representations are invertible.

4.17 Unimodular finite tensor category

A finite tensor category is **unimodular** if its distinguished invertible object α is the unit 1.

Example: In the category $\mathrm{Rep}(G)$, the 1-dimensional representations are invertible.

4.18 Indecomposable projective objects in a finite tensor category

Let $\mathcal{C}$ be a finite multitensor category over a field k. Let $\{X_i\}_{i\in I}$ be the set of isomorphism classes of simple objects in $\mathcal{C}$, and let P_i be the projective cover of X_i.

Then for any object $Z \in \mathcal{C}$, we have the following isomorphisms **in** $\mathcal{C}$:

1. $P_i \otimes Z \cong \bigoplus_{j,k\in I} N^i_{k,j^*}[Z : X_j]P_k$
2. $Z \otimes P_i \cong \bigoplus_{j,k\in I} N^i_{*j,k}[Z : X_j]P_k$

where:

- $[Z : X_j]$ is the multiplicity of the simple object X_j in the Jordan-Hölder series of Z.
- j^* and ${}^*j$ denote the indices for the right dual X_j^* and the left dual ${}^*X_j$, respectively.
- $N^c_{ab} := [X_a \otimes X_b : X_c]$ is the fusion coefficient, i.e., the multiplicity of X_c in the Jordan-Hölder series of $X_a \otimes X_b$.

Detailed Proof of Formula 1 We want to prove that:

$$P_i \otimes Z \cong \bigoplus_{j,k\in I} N^i_{k,j^*}[Z : X_j]P_k.$$

Step 1: The object $P_i \otimes Z$ is projective.

In a tensor category, the tensor product of a projective object with any other object is projective. Since P_i is projective by definition, the object $P_i \otimes Z$ is also a projective object in $\mathcal{C}$.

Step 2: Decomposition of projective objects.

Every projective object in a finite abelian category (like $\mathcal{C}$) can be uniquely decomposed (up to isomorphism and permutation) into a direct sum of indecomposable projective objects. The indecomposable projectives in $\mathcal{C}$ are precisely $\{P_k\}_{k\in I}$.

More precisely, there is a bijection between simple objects of $\mathcal{C}$ and indecomposable projective objects of $\mathcal{C}$, the bijection being $X_i \mapsto P_i$ and $P \mapsto P/\mathrm{rad}(P)$.

Therefore, we can write an isomorphism in $\mathcal{C}$:

$$P_i \otimes Z \cong \bigoplus_{k\in I} m_k P_k,$$

where m_k are non-negative integers representing the multiplicity of P_k in the decomposition. Our goal is to find an explicit formula for m_k.

Step 3: Calculating the multiplicities m_k.

The multiplicity m_k can be determined by computing the dimension of the space of morphisms from the object to the corresponding simple object X_k. This relies on the property of projective covers that $\mathrm{Hom}_{\mathcal{C}}(P_l, X_k) \cong k$ if $l = k$ and is 0 otherwise. More generally:

$$\dim \mathrm{Hom}_{\mathcal{C}}\left(\bigoplus_{l\in I} m_l P_l, X_k\right) = \sum_{l\in I} m_l \dim \mathrm{Hom}_{\mathcal{C}}(P_l, X_k) = \sum_{l\in I} m_l \delta_{lk} = m_k$$

Therefore, we can calculate m_k as:

$$m_k = \dim \mathrm{Hom}_{\mathcal{C}}(P_i \otimes Z, X_k)$$

Step 4: Using Tensor-Hom Adjunction.

The tensor functor $- \otimes Z$ has a right adjoint, which is $- \otimes Z^*$, where Z^* is the right dual of Z. This gives the following natural isomorphism of vector spaces (the "adjunction of rigidity"):

$$\mathrm{Hom}_{\mathcal{C}}(P_i \otimes Z, X_k) \cong \mathrm{Hom}_{\mathcal{C}}(P_i, X_k \otimes Z^*)$$

Combining this with the previous step, we get:

$$m_k = \dim \mathrm{Hom}_{\mathcal{C}}(P_i, X_k \otimes Z^*)$$

Step 5: Using the Defining Property of Projective Covers.

The projective cover P_i of the simple object X_i has the fundamental property that for any object $M \in \mathcal{C}$:

$$\dim \mathrm{Hom}_{\mathcal{C}}(P_i, M) = [M : X_i]$$

This means the dimension of the Hom space from P_i to M counts the number of times the simple object X_i appears in the composition series of M.

Let's apply this property with $M = X_k \otimes Z^*$:

$$m_k = [X_k \otimes Z^* : X_i]$$

as desired.

5 Module categories

5.1 Definition

Let $\mathcal{C} = (\mathcal{C}, \otimes, \mathbf{1}, a, l, r)$ be a monoidal category.

A **left module category** over $\mathcal{C}$ is a category $\mathcal{M}$ equipped with:

- a bifunctor $\otimes : \mathcal{C} \times \mathcal{M} \to \mathcal{M}$,
- a natural isomorphism (module associativity constraint)

$$m_{X,Y,M} : (X \otimes Y) \otimes M \xrightarrow{\sim} X \otimes (Y \otimes M),$$

- a natural isomorphism (unit constraint)

$$l_M : \mathbf{1} \otimes M \xrightarrow{\sim} M,$$

satisfying two coherence conditions:

- **Pentagon axiom:** the diagram for all $X, Y, Z \in \mathcal{C}$, $M \in \mathcal{M}$,

$$\begin{aligned} ((X \otimes Y) \otimes Z) \otimes M &\xrightarrow{a_{X,Y,Z} \otimes \mathrm{id}_M} (X \otimes (Y \otimes Z)) \otimes M \\ &\xrightarrow{m_{X,Y \otimes Z,M}} X \otimes ((Y \otimes Z) \otimes M) \\ &\xrightarrow{\mathrm{id}_X \otimes m_{Y,Z,M}} X \otimes (Y \otimes (Z \otimes M)) \\ &= (X \otimes Y) \otimes (Z \otimes M) \\ &\xleftarrow{m_{X,Y,Z \otimes M}} ((X \otimes Y) \otimes Z) \otimes M \end{aligned}$$

- **Triangle axiom:** for all $X \in \mathcal{C}, M \in \mathcal{M}$,

$$(X \otimes \mathbf{1}) \otimes M \xrightarrow{m_{X,\mathbf{1},M}} X \otimes (\mathbf{1} \otimes M) \xrightarrow{\mathrm{id}_X \otimes l_M} X \otimes M$$

  equals the map

$$(X \otimes \mathbf{1}) \otimes M \xrightarrow{r_X \otimes \mathrm{id}_M} X \otimes M$$

This categorifies the notion of a module over a monoid.

There is a bijective correspondence between:

- left $\mathcal{C}$-module category structures on $\mathcal{M}$, and
- monoidal functors $F : \mathcal{C} \to \mathrm{End}(\mathcal{M})$,

where $\mathrm{End}(\mathcal{M})$ is the monoidal category of endofunctors of $\mathcal{M}$ with composition.

If $\mathcal{C}$ is rigid, then for all $X \in \mathcal{C}, M, N \in \mathcal{M}$ there is a canonical isomorphism:

$$\mathrm{Hom}_{\mathcal{M}}(X^* \otimes M, N) \cong \mathrm{Hom}_{\mathcal{M}}(M, X \otimes N)$$

natural in all variables.

5.2 $\mathcal{C}$-module functor

Let $\mathcal{M}, \mathcal{N}$ be two left $\mathcal{C}$-module categories. A **$\mathcal{C}$-module functor** is:

- a functor $F : \mathcal{M} \to \mathcal{N}$,
- a natural isomorphism

$$s_{X,M} : F(X \otimes M) \xrightarrow{\sim} X \otimes F(M),$$

  natural in $X \in \mathcal{C}, M \in \mathcal{M}$, such that for all $X, Y \in \mathcal{C}, M \in \mathcal{M}$ the diagram

$$\begin{aligned} F((X \otimes Y) \otimes M) &\xrightarrow{F(m_{X,Y,M})} F(X \otimes (Y \otimes M)) \\ &\xrightarrow{s_{X,Y\otimes M}} X \otimes F(Y \otimes M) \\ &\xrightarrow{\mathrm{id}_X \otimes s_{Y,M}} X \otimes (Y \otimes F(M)) \\ &= (X \otimes Y) \otimes F(M) \\ &\xleftarrow{s_{X\otimes Y,M}} F((X \otimes Y) \otimes M) \end{aligned}$$

  commutes.

5.3 Module category over a multitensor category

- A **module category** over a multitensor category $\mathcal{C}$ is a locally finite abelian category $\mathcal{M}$ over a field k, with a $\mathcal{C}$-module category structure where the action functor

$$\otimes : \mathcal{C} \times \mathcal{M} \to \mathcal{M}$$

  is bilinear and exact in the first variable.

- Let $\mathrm{Lex}(\mathcal{M}, \mathcal{M})$ denote the category of left exact endofunctors of $\mathcal{M}$. Then there is a bijection between $\mathcal{C}$-module structures on $\mathcal{M}$ and tensor functors

$$F : \mathcal{C} \to \mathrm{Lex}(\mathcal{M}, \mathcal{M})$$

- The direct sum $\mathcal{M}_1 \oplus \mathcal{M}_2$ of two $\mathcal{C}$-module categories is again a $\mathcal{C}$-module category.
- A $\mathcal{C}$-module category $\mathcal{M}$ is *indecomposable* if it is not equivalent to a nontrivial direct sum of module categories.

Example: If $\mathcal{C}$ is a multitensor category, then $\mathcal{C}$ is a $\mathcal{C}^{env} := \mathcal{C} \boxtimes \mathcal{C}^{\mathrm{op}}$-module category with

$$(X \boxtimes Y) \otimes Z := X \otimes Z \otimes Y.$$